

21 2026-03-11 | Week 09 | Lecture 21

21.1 Normal random variables

This section is based on chapter 4.3.

Definition 91 (Normal / Gaussian Distribution). A random variable X is said to be **normally distributed** with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ if it has pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$$

defined for all $x \in \mathbb{R}$. We abbreviate this by writing $X \sim \mathcal{N}(\mu, \sigma^2)$. If $X \sim \mathcal{N}(0, 1)$, we say X has **standard normal distribution**. The parameter σ is called the **standard deviation** of X , and is the positive square root of the variance.

Remark 92 (Notation).

- It is a common convention to denote a standard normal random variable with the letter Z .
- random variables/distributions are also called **Gaussian** random variables/distributions.

Example 93. Normal distributions are the classic “bell curve” distributions. Whenever a random quantity is the result of adding up many small, independent random effects, we expect it to be approximately normally distributed.

- A classic example is human height, which is influenced by the combined effects of thousands of genes together with environmental factors.
- Another example is repeated measurements of the same physical quantity (e.g., if we all went out and measured the temperature outside tomorrow at midday), this is because the total error is the sum of many small independent sources of noise (e.g., instrument precision, time, location, shade, wind, etc).

End of Example 93. $\square$

Remark 94 (Standardization). If $Y \sim \mathcal{N}(\mu, \sigma^2)$ then the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a standard normal random variable (i.e., $Z \sim \mathcal{N}(0, 1)$). The transformation

$$X \mapsto \frac{X - \mu}{\sigma}$$

in which we subtract off the mean of X and then divide by its standard deviation is called **standardization**.

Remark 95 (The Empirical Rule). The **Empirical Rule** (also called the **68-95-99.7 rule**; see textbook section 1.3) is summarized by the following figure, for normal distributions:

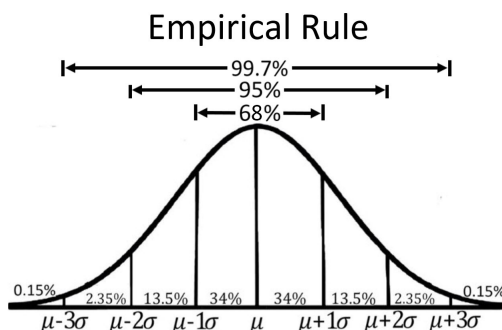


Image source: <https://andymath.com/wp-content/uploads/2019/12/empirical-rule-normdist2.jpg>

Remark 96. Note that μ determines where the peak of the bell curve is located, and σ determines how wide or narrow it is.

The next example shows how we can use standardization to do calculations:

Example 97. Due to variation in the manufacturing process, the peak power consumption X of the latest Nvidia GPU varies slightly from chip to chip. Assume the power consumption is normally distributed $X \sim \mathcal{N}(\mu, \sigma^2)$. What is the probability that X is within one standard deviation of its mean?

$$\begin{aligned}
 \mathbb{P}[\mu - \sigma \leq X \leq \mu + \sigma] &= \mathbb{P}\left[\frac{(\mu - \sigma) - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{(\mu + \sigma) - \mu}{\sigma}\right] \\
 &= \mathbb{P}\left[\frac{(\mu - \sigma) - \mu}{\sigma} \leq Z \leq \frac{(\mu + \sigma) - \mu}{\sigma}\right] && \text{by Remark 94} \\
 &= \mathbb{P}[-1 \leq Z \leq 1] \\
 &= \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
 &\approx .68 && \text{using the figure in Remark 95}
 \end{aligned}$$

So if $\mu = 700$ watts and $\sigma = 35$ watts (realistic values for the Nvidia's new \$30,000 H100 GPU), we have

$$\mathbb{P}[665 \leq X \leq 735] \approx 0.68.$$

In other words, about 68% of chips will have peak power consumption between 665 and 735 watts.

End of Example 97. $\square$